

Treatment of acute lymphoblastic leukaemia with the second generation of CD19 CAR-T containing either CD28 or 4-1BB

Shiqi Li,¹ Jiasi Zhang,² Meiling Wang,¹
Gang Fu,² Yunyan Li,¹ Li Pei,²
Zhouxing Xiong,¹ Dabing Qin,²
Rui Zhang,¹ Xiaobo Tian,² Zhihao Wei,¹
Run Chen,² Xuejiao Chen,¹ Jia Wan,²
Jun Chen,¹ Xia Wei,² Yanmin Xu,¹
Pei Zhang,² Ping Wang,² Xi Peng,²
Sainan Yang,² Junjie Shen,¹ Zhi Yang,¹
Jieping Chen² and Cheng Qian¹ 

¹Centre of Biological Therapy, Southwest Hospital, Third Military Medical University, and ²Centre of Haematology, Southwest Hospital, Third Military Medical University, Chongqing, China

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Correspondence: Prof. Cheng Qian, Centre of Biological Therapy, Southwest Hospital, Third Military Medical University, Chongqing, China.

E-mail: cqian8634@gmail.com
and

Prof. Jieping Chen, Centre of Haematology, Southwest Hospital, Third Military Medical University, Chongqing, China.

E-mail: chenjpxn@163.com

Abstract

T cells modified with anti-CD19 chimeric antigen receptor (CAR) containing either CD28 or 4-1BB (also termed TNFRSF9, CD137) costimulatory signalling have shown great potential in the treatment of acute lymphoblastic leukaemia (ALL). However, the difference between CD28 and 4-1BB costimulatory signalling in CAR-T treatment has not been well elucidated in clinical trials. In this study, we treated 10 relapsed or refractory ALL patients with the second generation CD19 CAR-T. The first 5 patients were treated with CD28-CAR and the other 5 patients were treated with 4-1BB CAR-T. All the 10 patients were response-evaluable. Three patients achieved complete remission and 1 patient with extramedullary disease achieved partial response after CD28-CAR-T treatment. In the 4-1BB CAR-T treatment group, 3 patients achieved complete remission. Furthermore, FLT-3 ligand (FLT3LG) was highly correlated with response time and may serve as a prognosis factor. No severe adverse events were observed in these 10 treated patients. Our study showed that both CD28 CAR-T and 4-1BB CAR-T both worked for response but they differed in response pattern (peak reaction time, reaction lasting time and reaction degree), adverse events, cytokine secretion and immune-suppressive factor level.

Keywords: Chimeric antigen receptor-T therapy, CD19, CD28/4-1BB costimulatory signalling, acute lymphoblastic leukaemia.

T cell modified with a chimeric antigen receptor (CAR) is a promising technology in adoptive immunotherapy. Since its first construction (Gross *et al*, 1989), CAR-T has been considerably developed in recent years. The CAR endows T cells with high antigen affinity in a major histocompatibility complex (MHC)-independent way, which is essential for cancer treatment. CD19 is a protein ubiquitously expressed in the B lymphocyte lineage, throughout the pre-B to mature B cell differentiation (Scheuermann & Racila, 1995). The expression of CD19 is stable even in neoplastic transformation, making it an ideal target for B cell malignancies. The second generation CAR-T specific for CD19 is a promising treatment for B cell malignancies and has been reported to be effective in pilot trials for relapsed and refractory B cell acute lymphoblastic leukaemia (ALL) (Davila *et al*, 2014; Maude *et al*, 2014; Lee *et al*, 2015; Turtle *et al*, 2016a), chronic lymphocytic leukaemia (CLL) (Brentjens *et al*, 2011; Porter *et al*,

2015) and non-Hodgkin lymphoma (NHL) (Kochenderfer *et al*, 2010; Savoldo *et al*, 2011; Turtle *et al*, 2016b).

CD28 and 4-1BB (also termed TNFRSF9, CD137) are the two most commonly used costimulatory signal molecules in recent clinical trials of the second generation CAR-T. Differences derived from the two signals have been illustrated in preclinical research (Milone *et al*, 2009; Long *et al*, 2015; Kawalekar *et al*, 2016). Published clinical trial results showed that CAR-T cells expressing a CD28 domain persisted for up to 3 months (Brentjens *et al*, 2013; Davila *et al*, 2014; Lee *et al*, 2015), while CAR-T cells with a 4-1BB domain persisted for up to 5 years and more than 6 months in most cases (Maude *et al*, 2014; Porter *et al*, 2015). However, no direct comparison of these two costimulatory signals in CD19 CAR-T cells for treatment of ALL patients has been investigated.

In this study, we carried out a phase I/II clinical trial of CD19 CAR-T to treat relapsed and refractory B cell

malignancies (clinicaltrials.gov: NCT02349698). Here we report the results of our first 10 ALL subjects after receiving infusion of CD19 CAR-T containing CD28 or 4-1BB. Our data shows the different effect of these two signals.

Methods

Clinical trial design and procedures

The trial was an open-labelled, non-randomized phase I/phase II clinical trial and was approved by the Ethics Committee of the Southwest Hospital of Third Military Medical University (Chongqing, China). The main purpose of this trial was to verify the safety and efficiency of different CAR-T cells targeting CD19 in the treatment of B cell malignancies. The study was registered at Clinicaltrials.gov (NCT02349698) and was performed according to the principles of the Declaration of Helsinki. All participants were well informed and volunteered to participate in this trial and provided written informed consent before enrolment. No commercial sponsor was involved in the study.

For those with bone marrow primary disease, tumour load and response to treatment were evaluated by bone marrow morphology and minimal residual disease negativity was defined as less than 0.01% marrow blasts by FCM. Complete response (CR) was less than 5% marrow blasts, absence of circulating blasts, and no extramedullary sites of disease with absolute neutrophil count $\geq 1.0 \times 10^9/l$ and platelet count $\geq 100 \times 10^9/l$. For those with extramedullary invasion, tumour load and response were judged by imaging examinations including computed tomography (CT), positron-emission tomography (PET)-CT and bone scanning. Inclusion and exclusion criteria were listed on Clinicaltrials.gov

(NCT02349698). Preconditioning was adopted as follow: fludarabine 25 mg/m²/day for 4 days followed by cyclophosphamide 900 mg/m²/day for 1 day. Patient 8 was pancytopenic before infusion and did not receive preconditioning therapy. The day before infusion was defined as day 0. CAR-T cells were administered by intravenous infusion in a split-dose manner during which 10%, 30% and 60% of total cells were separately infused on the first, second and third day. Escalating dose was not adopted because prior studies had proved that CD19 CAR-T was well tolerated at doses of $1 \times 10^6/kg$ to $1 \times 10^7/kg$ (Brentjens *et al*, 2011; Grupp *et al*, 2013), so doses of cells were determined according to the total amount of cells available after manufacture (manufacturing goal: $1 \times 10^6/kg$ to $1 \times 10^7/kg$). Specific infusion details are listed in Table I. Cells were infused for over one hour each day. Vital signs including breathing, temperature, pulse and blood pressure were monitored at real time after infusion. Blood tests were performed weekly for 1 month after infusion. Response to CAR-T was evaluated 20–30 days after infusion. Adverse events were graded by the National Cancer Institute Common Terminology Criteria for Adverse Events (CTCAE) version 4.03 (https://evs.nci.nih.gov/ftp1/CTCAE/CTCAE_4.03_2010-06-14_QuickReference_5x7.pdf).

Patient characteristics

All ten patients (three males and seven females) were enrolled at Southwest Hospital and had a Karnofsky score (KPS) above 80. The median age was 33 years (range: 18–59 years). Most subjects had experienced at least one cycle of standard chemotherapy and a mean of 2 (range: 1–3) relapses. Two subjects (Patients 4 and 10) had extramedullary invasions. Bone marrow blasts were verified before

Table I. Patient characteristics and summary of response.

Patient	Age (years)	Gender	Gene fusion	Duration of CR1 (months)	Relapses (n)	Marrow blasts (% of mononuclear) before treatment	Depletion therapy	CRS Grade*	Response and time since treatment (months)
1	45	Female	<i>ETV6/RUNX1</i>	24	2	2.5	FC	0	CR, MRD– (4)
2	59	Male	–	21	1	40	FC	2	CR, MRD– (8)
3	36	Male	<i>BCR/ABL1</i> <i>WT1</i>	7	2	6	FC	1	CR, MRD– (6)
4	31	Female	–	11	3	6.5	FC	2	PR (3)
5	18	Male	–	0	–	58.5	FC	2	NR
6	22	Female	–	7	1	52.5	FC	2	CR, MRD– (2)
7	20	Female	–	117	2	4.5	FC	0	CR, MRD– (8)
8	34	Female	<i>BCR/ABL1</i> (T315I)	24	1	0.5	None	1	NR
9	20	Female	–	5	2	22.5	FC	2	NR
10	41	Female	<i>KMT2A/AFF1</i>	3	3	<0.5	FC	1	CR (5+)

+, ongoing response at the time of writing; CR1, first complete remission; CRS, cytokine release syndrome; FC, fludarabine 25 mg/m²/day for 4 days followed by cyclophosphamide 900 mg/m²/day for 1 day; MRD, minimal residual disease as assessed by flow cytometry; NE, not evaluable; NR, no response; PR, partial response for extramedullary disease.

*classified by the Cancer Therapy Evaluation Program Common Terminology Criteria for Adverse Events Version 4.0.

infusion (median, 12.75%; range: 0.5–58.5%) and fusion genes were found in the blast cells of four patients. More details are shown in Table I.

CAR structure and generation of CD19 CAR-T cells

Details of lentivirus production and cell cultivation have been previously described (Zhang *et al*, 2017). For Patients 1–5, anti-CD19 scFv originating from FMC63 antibody was inserted into the pCDH lentiviral vector with a G4 h (IgG4 hinge), 28TM (CD28 transmembrane domain), and CD28 and CD3 zeta intracellular signalling domains. For Patients 6–10, anti-CD19 scFv was inserted into pCDH lentiviral vector with a G4 h (IgG4 hinge), 28TM (CD28 transmembrane domain), and 4-1BB and CD3 zeta intracellular signalling domains. Lentivirus was collected and purified from the supernatant of 293T cells.

For CAR-T cell culture, peripheral blood was collected from patients and peripheral blood mononuclear cells (PBMC) was isolated with Ficoll-Paque PLUS (GE Healthcare Life Sciences, Chalfont St Giles, Buckinghamshire, UK, 17-1440-02) on day 1. PBMC were activated by immobilized CD3 (magnetic-activated cell sorting [MACS], Miltenyi Biotec, Bergisch Gladbach, Germany); 170-076-116) and CD28 (MACS, Miltenyi Biotec, 170-076-117) antibodies. On day 2, T cells were transfected with lentiviral particles in 6-well plates (1×10^6 cells per well, 3 multiplicity of infection) with 8 µg/ml polybrene (107689; Sigma-Aldrich, St Louis, MO, USA). After viral transduction, T cells were stimulated and expanded by interleukin (IL)-2 (100 u/ml) for approximately 14 days to obtain sufficient cell dose. Quality control of cell products was performed by FCM. For the cell products, the CAR+%, proportion of T cells (CD3+), NK cells (CD16+CD56+), NKT cells (CD3+CD16+CD56+), CD4/8 ratio, T memory cells and checkpoint inhibitors (TIM-3, PD-1, LAG-3) on the T-cell surface were measured on the day before CAR-T infusion.

Flow cytometry detection

BD FACS Calibur and BD LSRFORTESSA were used for FCM detection (all Becton Dickinson [BD], San Jose, CA, USA), and BD FACS DiVa software and FlowJo were used for analysis. Recombinant CD19 with a His tag (Sino Biological Inc. (Beijing, China) 11880-H08H) or Protein L (Thermo Fisher Scientific (Waltham, Massachusetts, USA), 21189) were administered for CAR+%-specific detection. Antibodies specific for CD3 (BD Biosciences (San Jose, California, USA), 344818), CD4 (BD Biosciences, 300506), CD8 (BD Biosciences, 344740), CD14 (BD Biosciences, 555397), CD25 (BD Biosciences, 555433), CD45 (BD Biosciences, 560367), CD45RA (BD Biosciences, 304130), CD45RO (BD Biosciences, 304222), CD62L (BD Biosciences, 555544), CD127 (BD Biosciences, 562436) and HLA-DR (BD Biosciences, 555813) were applied. TIM-3 (BD Biosciences,

536422), PD-1 (BD Biosciences, 560908) and LAG-3 (BD Biosciences, 565616) were also detected in CD4+ and CD8+ cells. Lymphocyte subtypes were measured using the MultiT-EST IMK Kit (BD Biosciences, 340503). Isotype antibodies and untransduced T cells were used as controls.

Quantitative DNA copies

Blood samples were obtained following the trial procedure. Genomic DNA was extracted from the blood samples using a QIAamp DNA Blood Mini Kit (Qiagen, Duesseldorf, Germany, 51106) and following the protocol. We applied Taqman probes for real time polymerase chain reaction (qPCR) in an ABI QuantStudio 5 (Applied Biosystems, Foster City, CA, USA). Taqman primers were 5'-CTTCCCGTATGGCTTTCATTTTCTC-3' (Forward) and 5'-CGCCACGTTGCCTGACA-3' (Reverse). Probe was 5'-FAM-ACGGGCCACAACCTCT-3'. The Taqman method was performed in accordance with the published protocol (Kalos *et al*, 2011). *CDKN1A* was used as the control and produced a correction factor (CF) for correction of DNA copy number. Healthy donor DNA was used as negative control and to define the pre-infusion acceptance ranges of qPCR method. The lower limit of quantification (LLOQ) was determined at 1 copy ng genomic DNA.

Cytokines detection

Plasma samples were stored at -80°C and were analysed in batches using a 15-plex human cytokine magnetic bead panel (Milliplex, Darmstadt, Germany, SPR372; with the MAGPIX instrument (Luminex, Austin, TX, USA). Levels of the cytokines IL-2, IL-6, IL-10, FLT3LG, Fractalkine, GM-CSF, IFN- γ , TNF- α , IL-2R α , IL-1RA, MIG (CXCL9), IP-10, MCP-1, MIP-1 α and MIP-1 β were measured. Samples were undiluted, and each was assayed in duplicate. Data analysis was conducted following the related protocol and algorithm.

Statistical analysis

Descriptive statistics were reported as median and ranges. Comparison was made by Mann–Whitney test or Wilcoxon matched pairs test. Correlation was analysed by Spearman correlation and the correlation coefficient was calculated. Statistical significance was defined as $P < 0.05$.

Results

Characteristics of ALL patients and CAR-T cells

A total of 13 patients were enrolled in this study. Three patients provided informed consent for the protocol but did not receive the CAR T-cell infusion due to disease progression. A total of 10 patients received infusion and all the patients are response-evaluable. Patients were treated with the standard procedure for treatment and management of

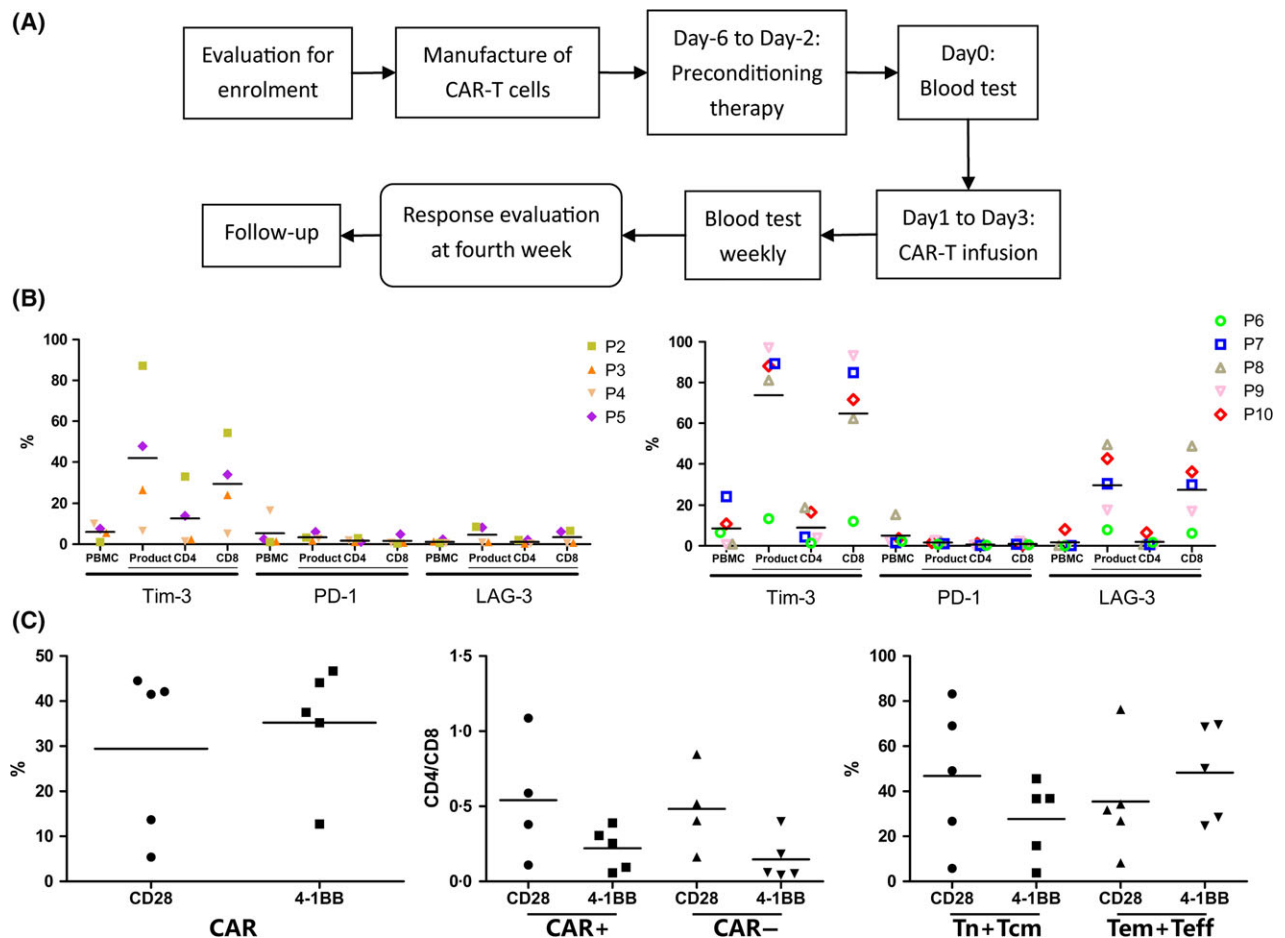


Fig 1. Procedure of clinical trial and characteristics of CAR-T cells. (A) Trial schedule. (B) Expression of Tim-3, PD-1 and LAG-3 on T cells before cultivation (PBMC) and after cultivation (products). CAR, chimeric antigen receptor; PBMC, peripheral blood mononuclear cells; Tcm, central memory T cell; Teff, effector T cell; Tem, effector memory T cell; Tn, native T cell. [Colour figure can be viewed at wileyonlinelibrary.com]

CD19 CAR-T therapy in ALL (Fig 1A). CAR-T cells containing CD28 were infused to the first 5 patients (CD28 group) and CAR-T cells containing 4-1BB were given to the other 5 patients (4-1BB group). A median $0.62 \times 10^6/\text{kg}$ (range, $0.1 \times 10^6/\text{kg}$ to $9.79 \times 10^6/\text{kg}$) CAR-T cells were infused. The CAR-T cells were manufactured over a period of about 14 days and the phenotypes of cell product were measured by flow cytometry (FCM) 1 day before infusion. The expression of Tim-3 and LAG-3 on T cells was increased after cultivation in both the CD28 group and 4-1BB group, when compared with pre-transduction PBMC. However, the increase was not statistically significant, possibly due to limited sample volume (Fig 1B). PD-1 (PDCD1) expression was not affected after cultivation in both groups (Fig 1B). The median CD3+ T cell content was 98.3% (range, 94–99.7%). Memory T cells were identified as CD45RA, CD45RO, and CD62L under the CD3+ group. Native T (Tn)/Stem cell-like T (Tscm; CD45RA+CD45RO-CD62L+), central memory T (Tcm; CD45RA-CD45RO+CD62L+), effector memory T (Tem; CD45RA-CD45RO+CD62L-) and effector T (Teff; CD45RA+CD45RO-CD62L-) cells were measured in

product cells. There was no difference between the CD28 group and the 4-1BB group in the CAR transfection rate, proportion of CD4/CD8 in CAR+ group and CAR- group and proportion of memory phenotypes (Fig 1C). Infusion details are listed in Table II.

Response evaluation

The treatment process is shown in Fig 1A. Seven of the 10 evaluable patients showed response to treatment. In the CD28 group, 3 patients (Patients 1, 2 and 3) achieved complete remission (CR) and 1 patient (Patient 4) with left femur invasion achieved partial response (PR). In the 4-1BB group, 3 patients (Patients 6, 7 and 10) achieved CR (Fig 2A). Of the 6 patients who achieved CR, 5 relapsed at 2–8 months (median, 6 months) after infusion; the relapsed leukaemia cells were verified CD19+ by FCM. No relapses with CD19- ALL cells were observed (Fig 2B). At the time of writing, one patient had maintained CR for over 5 months without relapse (Fig 2A). Three patients showed no response: Patient 5 in the CD28 group had a high level of marrow

Table II. Infused cell details.

Patient	Total cells infused		Transduced %	CAR-T cells infused			CD4/CD8 ratio		Phenotype of infused T cells			
	Dose $\times 10^7$	CD 3%		Total $\times 10^7$	Dose $\times 10^6/\text{kg}$	Costimulatory signal	CAR+	CAR–	Tn/Tscm %	Tcm %	Tem %	Teff %
1	3.3	97.5	41.5	1.33	0.27	CD28	0.163	0.163	3.3	2.45	62.7	13.55
2	155	94	13.7	19.96	3.7	CD28	1.087	0.517	10.63	16.08	28.19	3.55
3	10.1	98.6	44.5	4.43	0.63	CD28	0.11	0.163	66.49	2.58	16.12	10.79
4	11.7	99	42.1	4.88	0.61	CD28	0.38	0.404	72.68	10.55	6.938	1.234
5	12.2	94.4	5.4	0.62	0.1	CD28	0.588	0.846	33	16.16	27.15	7.18
6	5.31	99.7	37.5	1.99	0.31	CD137	0.39	0.182	29.42	7.38	2.8	25.65
7	9.5	98.6	35.2	3.3	0.66	CD137	0.095	0.042	3.15	33.52	24.18	0.55
8	139	98	46.7	63.61	9.79	CD137	0.306	0.052	1.87	1.90	52.40	16.13
9	11.7	99.7	44.1	5.14	1.03	CD137	0.057	0.059	0.62	44.89	49.81	0.28
10	6.95	95.2	12.72	0.84	0.17	CD137	0.255	0.398	14.42	1.45	8	61.52

CAR, chimeric antigen receptor; Tcm, central memory T cell; Teff, effector T cell; Tem, effector memory T cell; Tn, native T cell; Tscm, stem cell-like T cell.

blasts (58.5%) while Patients 8 and 9 in the 4-1BB group had low marrow blasts (0.5% and 22.5%).

CD19+ B lymphocytes in peripheral blood were measured by FCM within 1 week before CAR-T infusion (median, 0.78%; range, 0.0–17%) and were regularly measured after CAR-T infusion according to protocol (Fig 2B). The responding patients had undetectable CD19+ cells in peripheral blood (median 0.0%; range, 0.0–0.1%) around 1 month after infusion, whereas, in non-responders, peripheral blood CD19+ cells rapidly increased within 2 weeks after infusion (Patients 5 and 8) or increased during the third week after 2 weeks of undetectable levels (Patient 9). CD19+ cell recovery (>3%) occurred at 2–4 months after infusion in all the responders except for Patient 10, who was still in B cell aplasia at the last follow-up (over 5 months).

Our study showed that CAR-T containing CD28 or 4-1BB both worked for extramedullary invasions. Patient 4 was diagnosed with ALL by marrow morphology and extramedullary disease in the left femur was detected by whole body bone scan, which was verified as ALL by bone biopsy. The phenotype of her malignant cells was verified as CD19+CD10+HLA–DR+CD15dimCD34–CD20– by FCM. She achieved CR with the VDLD regimen (vincristine, daunorubicin, L-asparaginase, dexamethasone) and maintained CR through consolidations including CHOP (cyclophosphamide, daunorubicin, vincristine, prednisolone), CAM (cyclophosphamide, cytarabine, 6-mercaptopurine), mM (methotrexate, 6-mercaptopurine) and HR-1 (vincristine, dexamethasone, cyclophosphamide, cytarabine, L-asparaginase) for a year until the pain re-appeared in her left femur with normal bone marrow. Subsequent HR-2 regimen (dexamethasone, cyclophosphamide, L-asparaginase, vindesine, ifosfamide, daunorubicin) only relieved the pain for a month. This patient was enrolled in our trial and achieved PR 1 month after CAR-T infusion evaluated by bone scan (Fig 2C). The response lasted for 3 months until another invasion was verified on the right femur by magnetic resonance imaging. Patient 10 received

allogeneic transplantation 5 months after her diagnosis and achieved 100% chimerism rate. CR was continued in her marrow but positron emission tomography (PET) showed multiple metastasis 10 months after transplantation. The metastasis was verified as invasion of B-ALL by biopsy. Before CAR-T infusion, the metastases in her breasts (bilateral; Fig 2D top-left), right ilium, pelvic lymph nodes (Fig 2D bottom-left), left femur and vertebrae were evaluated by PET. The patient then received donor-derived but autologously-collected CAR-T cells, with most metastases disappearing one month after infusion (Fig 2D top-right and bottom-right) and reduced residues in her right breast and left femur. These residues continued to shrink and all residues had disappeared by 5 months after infusion.

Expansion and persistence of CAR-T Cell in peripheral blood

CAR-T cells in peripheral blood were detected weekly by FCM (Fig 3A) and qPCR (Fig 3B). Protein L, shown to be a good reagent for the detection of CAR expression by FCM (Zheng *et al*, 2012), was used in our study. For all the responders except for Patient 1, CAR-T cells were detectable (>5% in T cells) in peripheral blood by FCM within 1 month after infusion. The peak of CAR was higher in the 4-1BB group (median, 29.44%; range, 25.6–33.4%) than in the CD28 group (median, 9.45%; range, 2.8–25.4%) among responders, although the difference was not statistically significant ($P = 0.0571$). Considering CAR copy numbers and the recovery of CD19+ B cells, the persistence of CAR-T cells over 2 months was only observed on Patient 10.

Adverse events

For the ten enrolled patients, CAR-T was well-tolerated and no adverse events were over grade 2 except for haematological changes induced by chemotherapy preconditioning.

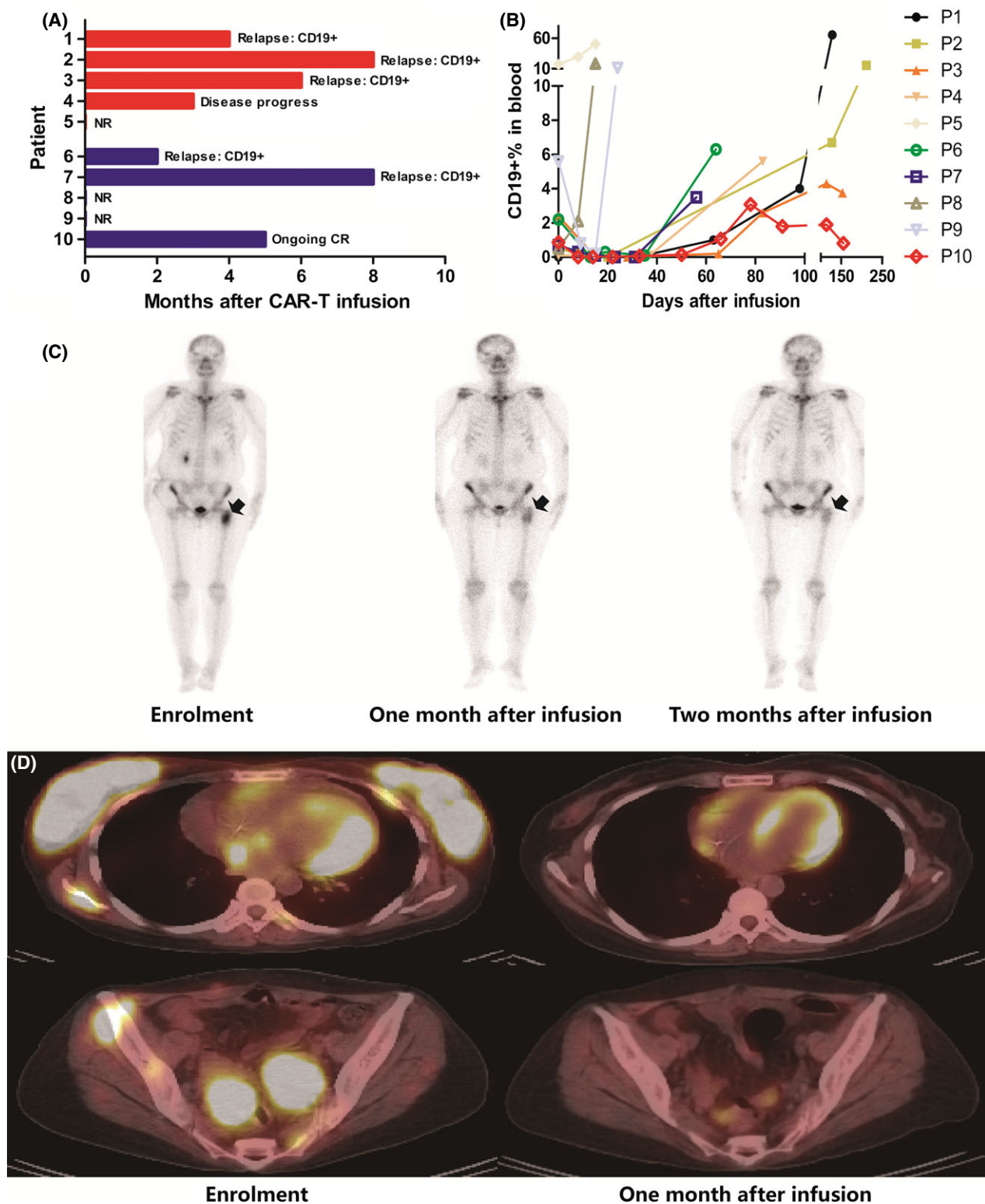


Fig 2. (A) Clinical outcomes of acute lymphoblastic leukaemia (ALL) patients treated with CD19 CAR-T cells. (B) Percentage of CD19+ cells in lymphocytes in blood detected by flow cytometry. (C) Bone scans of Patient 4 before CAR-T infusion (left) and 1 month (middle) and 2 months after infusion. The black arrow indicates the invasion of ALL on her left femur. (D) Positron emission tomography/computed tomography assessment of Patient 10 before CAR-T infusion (left) and 1 month after infusion (right). CAR, chimeric antigen receptor; CR, complete remission; NR, no response. [Colour figure can be viewed at wileyonlinelibrary.com]

Pancytopenia was the most common adverse event and all responders recovered within 2 months. Four patients in the CD28 group and four patients in the 4-1BB group had fever within 2 weeks after infusion, all of which were relieved by symptomatic treatments. The peak levels of C-reactive protein (CRP), IL-10 and IL-6 appeared earlier in the CD28 group (days 5–9) than in the 4-1BB group (days 9–15) (Fig 4A). Other adverse events, including fatigue, nausea, headache, myalgia, ostealgia and chill, were all slight and self-limiting. Grade 1 graft-versus-host disease (GVHD) induced by donor-derived CAR-T cells was observed 2 months after infusion in Patient 10, who achieved complete chimism after transplantation and showed no signs of GVHD before enrolment. Abnormalities, including hyperbilirubinaemia, elevated aminotransferases and oral ulcers, were controlled and reversed by administration of azathioprine and methotrexate.

Inflammatory factor levels, including CRP and IL-6, were detected in fresh plasma samples by routine laboratory examination. Cytokines including IL-6, IL-10, IL-1RA, IFN- γ , MIG, GM-CSF, Fractalkine, IP-10, TNF- α , MCP-1, MIP-1a, IL-2 and FLT3LG were measured in cryopreserved plasma samples by Milliplex assay. IL-6 was measured in both fresh and cryopreserved plasma. Spearman correlation revealed high consistency ($r = 0.7891$, $P < 0.0001$) between the two results, but Wilcoxon matched pairs test showed that the result of cryopreserved plasma was lower ($P < 0.0001$). This may be due to the degradation of cytokines from long-term frozen storage.

Cytokine and biochemical marker analysis

There was more TNF- α secretion in the 4-1BB group than in the CD28 group ($P = 0.0079$) (Fig 5A left). There was no significant difference between the CD28 group and the 4-1BB group with regard to peak secretion of FLT3LG, but was significant between the responders and non-responders ($P = 0.0167$, Fig 5A centre). The peak secretion of FLT3LG was highly correlated with the progression-free survival of responders ($r = 0.7878$, $P = 0.0088$) (Fig 5B right). A heat map of cytokine levels within 1 month after infusion is shown in Fig 5B. IL-6, IL-10, CRP, IL-1RA, IFN- γ and MIG showed high secretion peaks while there was no significant difference between the CD28 group and the 4-1BB group for these six factors.

Spearman correlation was used to analyze the relationship in cytokine secretion between the CD28 group and the 4-1BB group (Fig 5C). IL-6, IL-10, Fractalkine, IFN- γ , GM-CSF, IL-2 and MIP-1b were pairwise correlated both in the CD28 group and 4-1BB group, indicating these seven factors may follow the same secretion pattern regardless of costimulatory signals. IL-2RA, TNF- α , MIP-1a, IL-1RA, IP-10 and MIG showed the most discrepancies (9–13) in correlation with the other 15 cytokines between the CD28 group and the 4-1BB group, suggesting that these 6 cytokines may be secreted in a different pattern in the two signals. CRP, FLT3LG and MCP-1 were the only three factors which showed no correlation with most cytokines in both the CD28 and 4-1BB groups.

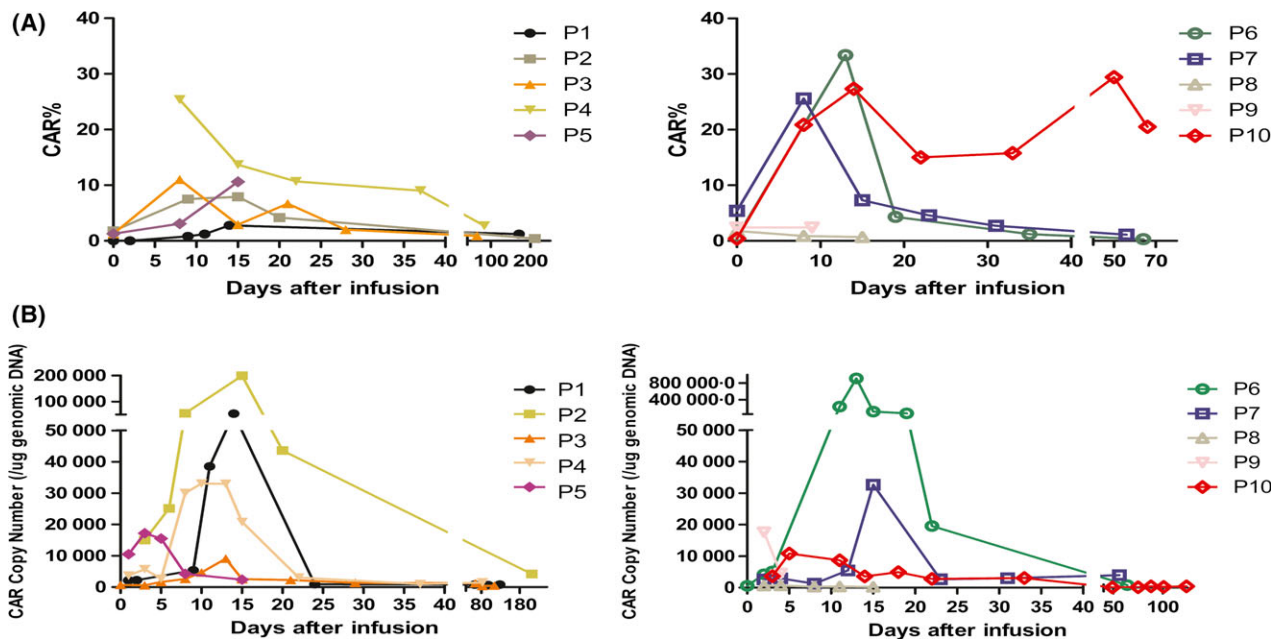


Fig 3. Detection of chimeric antigen receptor (CAR)-T cells in peripheral blood. (A) CAR+ percentage in the CD3+ lymphocytes measured by FCM. (B) Detection of DNA copies of CAR-T cells in peripheral blood by real time polymerase chain reaction. [Colour figure can be viewed at wileyonlinelibrary.com]

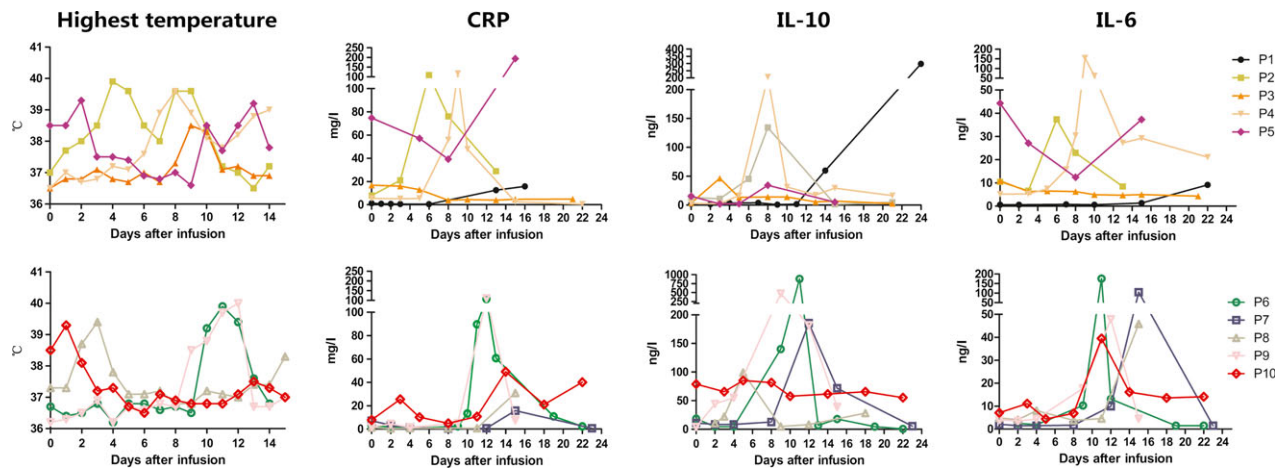


Fig 4. Highest temperature was recorded daily for those patients with fever after chimeric antigen receptor (CAR)-T infusion. Serum levels of C-reactive protein (CRP), interleukin (IL)-10 and IL-6 were measured at different time after CAR-T cell infusion. [Colour figure can be viewed at wileyonlinelibrary.com]

Detection of immune suppressive molecules

The expression of Tim-3, PD-1 and LAG-3 on T cells were measured by FCM on T cells pre-lentiviral transduction, post-transduction and post-infusion among all the patients except for Patient 1 (Table III). The expression level of these three markers was increased in most patients post-infusion in both the CD28 group and the 4-1BB group compared with T cells at enrolment (Fig 6A). However, only the increase of PD-1 ($P = 0.0317$) and LAG-3 ($P = 0.0317$) expression in the 4-1BB group was statistically significant. Around day 8 after infusion, the CD28 group showed a peak level of Tim-3 and the 4-1BB group showed a peak level of PD-1. There was higher PD-1 expression level on the T cells in the 4-1BB group (median, 29.75%; range, 6.8–91.2%) than in the CD28 group (median, 5.45%; range, 5.2–27.9%) but the difference was not significant ($P = 0.0667$). Spearman correlation analysis revealed that CAR expression was correlative with PD-1 expression in both the CD28 group and the 4-1BB group, while the correlation between CAR and Tim-3 expression only appeared in the CD28 group, and the correlation between CAR and LAG-3 expression was only shown in the 4-1BB group (Fig 6B). Tregs (CD4+CD25+CD127[−]) and myeloid-derived suppressor cells (CD45+CD14+HLA-DR[−]) in peripheral blood were also increased in some patients after infusion (Fig 6C).

Discussion

In our pilot trial, 10 patients with refractory or relapsed ALL were treated with second generation CAR-T cells using either CD28 or 4-1BB as costimulatory signal. In total, 7 patients showed response to CAR-T therapy and the response rate was close to published studies (Dai *et al*, 2015; Lee *et al*, 2015). Furthermore, our study showed that CD19 CAR-T containing either CD28 or 4-1BB both worked for extramedullary invasions, indicating that CAR-T is also suitable for

those with extramedullary invasions. However, the CD28 CAR-T cells only triggered a PR for one patient and there were no changes in the residue thereafter. The 4-1BB CAR-T cells exhibited ongoing anti-tumour response for one patient and eradicated the residues by 5 months after infusion. It would be interesting and meaningful to verify if 4-1BB CAR-T is more suitable for those with extramedullary invasions in a larger cohort of patients.

Most responders relapsed or progressed, regardless of whether the costimulatory signal was CD28 or 4-1BB. Post-CAR-T relapse is a great concern for researchers all around the world. According to published data, the relapse rate ranges from 26% to 100% among different studies (Maude *et al*, 2014; Lee *et al*, 2015; Turtle *et al*, 2016a; Chen *et al*, 2017). The mechanism behind this is complex and multifactorial. However, a consensus is developing that prolonged persistence of CAR-T cells may prevent relapse, particularly for relapse with CD19⁺ ALL cells (Maus & June, 2016). As CD19⁺ lymphocytes were targeted by CAR-T cells, the recovery of CD19⁺ lymphocytes indirectly indicated the disappearance of CAR-T cells. Maude *et al* (2014) suggested that CD19⁺ B cell recovery was a high risk indicator for relapse. In our study, CD19⁺ B cell recovery appeared in 6 responders at 2–4 months after infusion and these responders all relapsed or progressed with CD19⁺ leukaemia cells. Different from those with CD19[−] relapse in the presence of CAR-T, CD19⁺ relapse may be due to the short persistence of CAR-T cells *in vivo*.

There are numerous elements related to CAR-T persistence, of which the costimulatory signal and T cell subsets may be two major factors. In prior studies, CAR-T containing CD28 persisted for 1–3 months *in vivo* (Brentjens *et al*, 2011; Lee *et al*, 2015) while CAR-T containing 4-1BB persisted up to 5 years (Maude *et al*, 2014; Porter *et al*, 2015). Accumulated evidence attempted to explain why there was a difference in duration between CD28 and 4-1BB induced

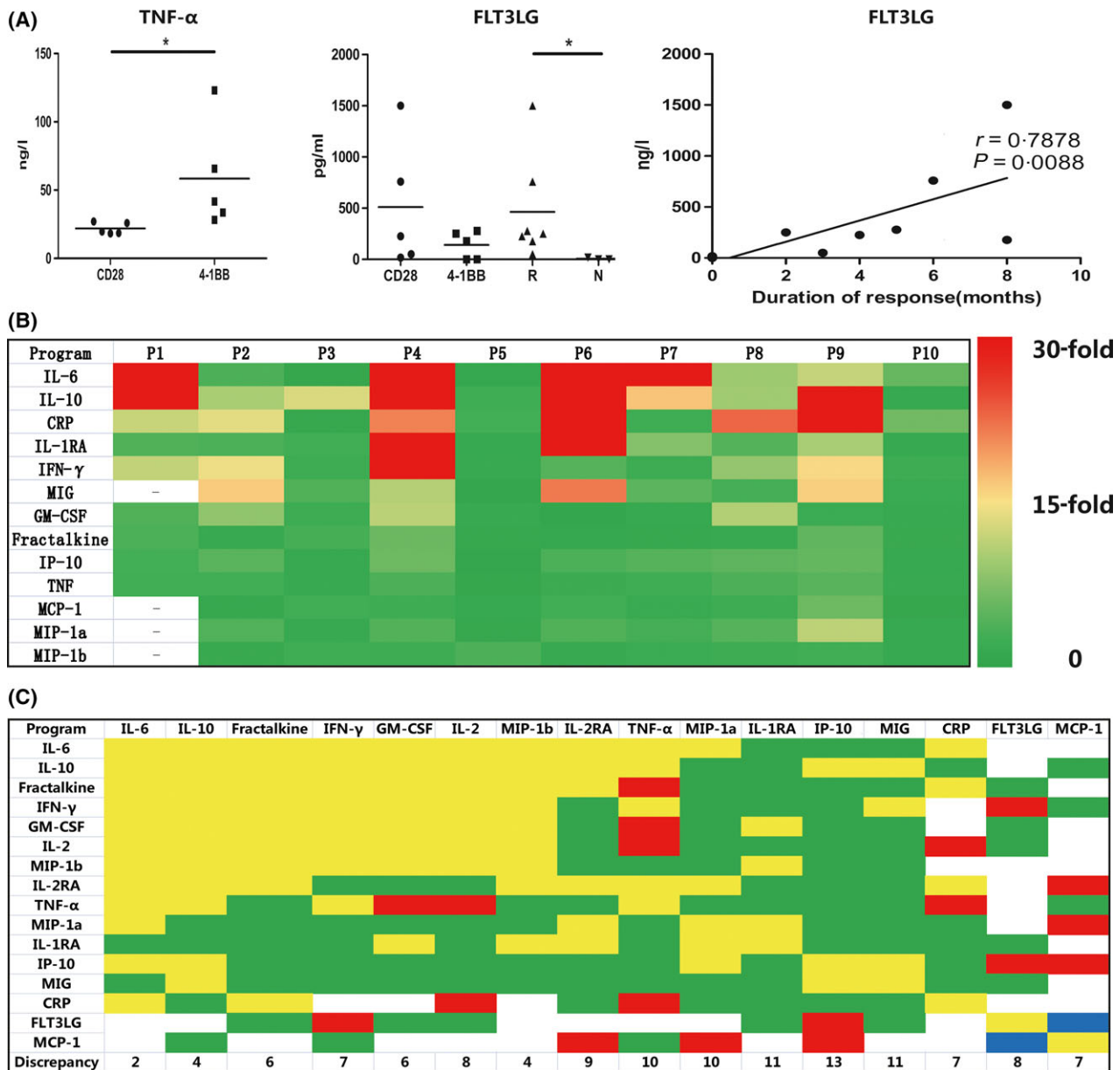


Fig 5. (A) Patients in the 4-1BB group showed higher secretion peak of TNF- α than in those in the CD28 group ($P = 0.0079$). There was no significant difference between the secretion peak of FLT3LG in the CD28 group and 4-1BB group but the difference was significant between responders and non-responders ($P = 0.0167$). The peak secretion of FLT3LG also correlated with response time ($r = 0.7878$, $P = 0.0088$). (B) Heat map of cytokine fold peak after chimeric antigen receptor (CAR)-T infusion. (C) Spearman correlation analysis between pairwise cytokines. The yellow blocks indicate that the two cytokines were positively correlated in both the CD28 group and the 4-1BB group. The red blocks indicate that the two cytokines were positively correlated in the CD28 group only. The green blocks indicate that the two cytokines were positively correlated in the 4-1BB group only. The blue blocks indicate that the two cytokines were negatively correlated in the 4-1BB group only. The white blocks indicate that the two cytokines did not correlate either in the CD28 group or in the 4-1BB group. [Colour figure can be viewed at wileyonlinelibrary.com]

activation (Zhang *et al*, 2007; Milone *et al*, 2009; Kawalekar *et al*, 2016). A group of T cell subsets was also found to correlate with *in vivo* CAR-T expansion and long term persistence (Xu *et al*, 2014; Singh *et al*, 2016; Bailey *et al*, 2017). In our trial the persistence of CAR-T cells was no more than a month in most patients despite of the costimulatory signal contained. The short persistence of CAR-T containing CD28 was consistent with previous studies, while the persistence in

4-1BB group differed. Barrett *et al* (2014) reported that compared with bead-stimulated cells, those generated with OKT3/IL2 exhibited worse disease control and shorter persistence in xenograft models of leukaemia. We postulate that the short persistence of our CAR-T cells containing 4-1BB may be due to the OKT3/IL-2 based CAR-T manufacturing method, which generates robust differentiated cells and few Tn/Tscm cells. In our ongoing study, patients are treated

Table III. Expression (%) of Tim-3, PD-1 and LAG-3 in the sample of T cells pre-lentiviral transduction, post-transduction and post-infusion.

Patient	Pre-transduction			Post-transduction			Post-infusion (peak)		
	Tim-3	PD-1	LAG-3	Tim-3	PD-1	LAG-3	Tim-3	PD-1	LAG-3
2	1.00	1.00	0.50	87.20	3.20	8.5	37.6	5.9	18.3
3	5.50	1.40	1.30	26.50	1.80	1.00	18.8	4.7	3.0
4	10.00	16.50	0.50	6.50	2.60	0.80	28.0	27.9	10.5
5	7.60	2.50	2.40	47.80	6.00	8.20	21.3	5.2	13.9
6	6.50	2.30	0.00	13.40	0.80	7.90	1.80	6.80	2.10
7	24.20	1.50	0.10	89.30	1.00	30.40	4.30	21.10	52.50
8	0.90	15.40	0.20	81.10	2.10	49.50	2.20	38.40	1.60
9	0.60	2.50	0.30	97.10	2.70	17.60	28.10	15.10	11.00
10	10.80	3.60	8.00	88.12	1.60	42.61	31.32	91.20	73.92

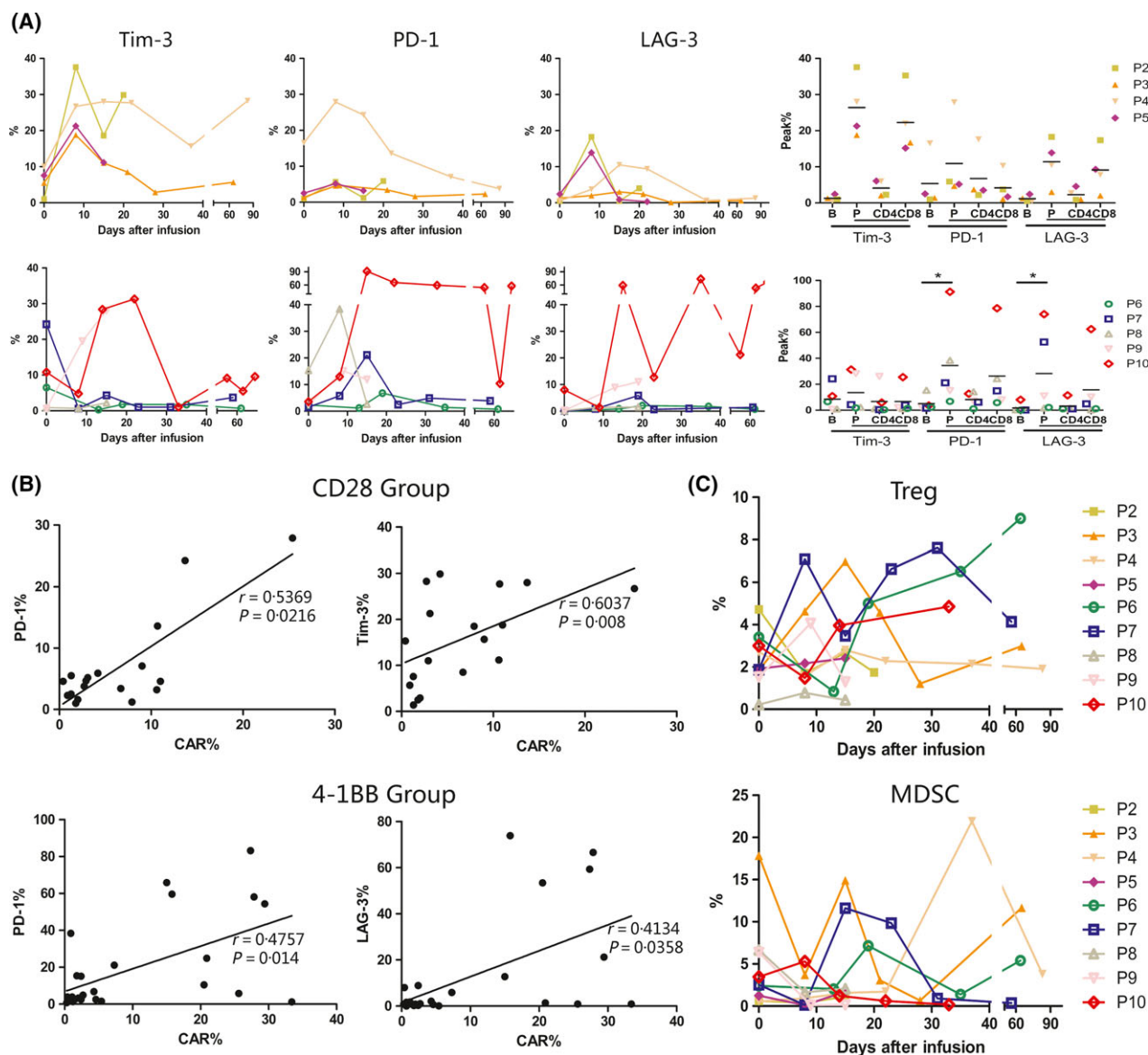


Fig 6. (A) Expression of Tim-3, PD-1 and LAG-3 on T cells in blood after CAR-T infusion. (B) Spearman correlation analysis revealed that CAR expression correlates with PD-1 expression both in the CD28 group and 4-1BB group, while correlates with Tim-3 expression only in the CD28 group and correlates with LAG-3 expression only in the 4-1BB group. (C) Detection of Treg and MDSC cells in blood. B, before infusion; CAR, chimeric antigen receptor; MDSC, myeloid-derived suppressor cells; P, Peak after infusion. [Colour figure can be viewed at wileyonlinelibrary.com]

with CAR-T cells generated with CD3/CD28 beads and the differences are being observed.

Adverse events were mild for all subjects enrolled during our trial. This may be due to the low tumour burden of most patients before infusion. The direct relationship between tumour burden and severity of adverse event had been reported by many studies and verified by most researchers. There was only one patient in each group with >50% marrow blast (Patient 5 in CD28 group: marrow blasts, 58.5% of karyocytes; Patient 6 in the 4-1BB group: marrow blasts, 52.5% of karyocytes) enrolled in our trial and the patient in the CD28 group showed no response to CAR-T infusion. This was consistent with what was reported, that high tumour load may invalidate CAR-T cells through antigen-specific activation driven by CD28 (Brentjens *et al*, 2011; Lee *et al*, 2015).

GVHD induced by donor-derived CAR-T cell infusion was observed in Patient 10. The occurrence of GVHD after infusion of donor-derived CAR-T cells was rare in previous studies. Dai *et al* (2015) reported two cases of GVHD induced by donor-derived CAR-T infusion and progression of malignancy after anti-GVHD therapy with short-term steroid treatment. CAR-T cells were still detectable in the blood of this patient when GVHD occurred and the GVHD was not severe, so steroids were not administered. Azathioprine and methotrexate controlled and reversed the GVHD well and did not affect the survival of CAR-T cells. The patient obtained an ongoing anti-tumour response and eradication of all residues by 5 months after CAR-T infusion.

In contrast to our previous CAR-T study in solid tumours (Zhang *et al*, 2017), immunosuppressive markers were increased in most patients after cultivation and after infusion. Spearman correlation analysis revealed the relationship between CAR expression and immunosuppressive markers. It has been known that the expression of immunosuppressive markers increases on the surface of activated T cells but we

are not sure if this correlates with CAR-T behaviour *in vivo*. In Patient 10, the expression of Tim-3, PD-1 and LAG-3 on T cells was high and CAR-T cells were detectable in her blood for over 2 months. These markers may be an indicator for response measurement.

Taken together, our study showed that both the CD28 CAR-T and the 4-1BB CAR-T worked for response but differed in response pattern (peak reaction time, reaction lasting time and reaction degree), adverse events, cytokine secretion and immune-suppressive factor level.

Authors' contributions

Jieping Chen, Zhi Yang and Cheng Qian designed the research study. Shiqi Li, Jiasi Zhang, Gang Fu, Li Pei, Dabing Qin, Xiaobo Tian, Run Chen, Jia Wan, Xia Wei, Pei Zhang, Ping Wang, Xi Peng and Sainan Yang performed the research. Meiling Wang and Zhouxing Xiong analysed the data. Yunyan Li and Rui Zhang cultivated cells for clinical use. Junjie Shen, Zhihao Wei, Xuejiao Chen, Jun Chen and Yanmin Xu performed preclinical study and contributed essential reagents and tools. Shiqi Li wrote the paper.

Competing interests

The authors declare that they have no competing interests.

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